

- 7 (a) nuclei with the same number of protons
and a different number of neutrons B1
B1 [2]
- (b) (i) (mass + energy) (taken together) is conserved (B1)
momentum is conserved (B1)
one point required max. 1 B1 [1]
- (ii) $a = 1$ and $b = 0$ B1
 $x = 56$ B1
 $y = 92$ B1 [3]
- (c) proton number = 90 B1
nucleon number = 235 B1 [2]